EIN 4343 Inventory and Supply Chain Systems

Prof. Elif Akçalı

Copyright © 2003-2026 Elif Akçalı

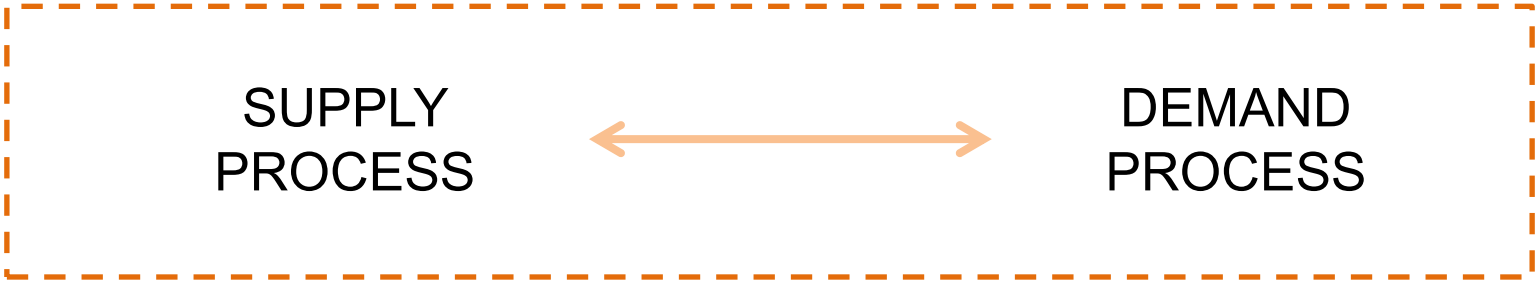
EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INTRODUCTION

Copyright © 2003-2026 Elif Akçalı

Why is there a supply chain?

- A product (or a service) satisfies a need or a want.



- Supply chain encompasses all of the agents that work toward matching the supply process with the demand process.

Whether a product (or a service) satisfies a need (e.g., food) or a want (e.g., smart phone) influences (1) the characteristics of the demand process and (2) how the supply process should be designed to match the demand process.

Copyright © 2003-2026 Elif Akçalı

What is a supply chain?

- Consists of stages involved, directly or indirectly, in fulfilling a customer need/want
  - Manufacturers, suppliers, transporters, warehouses, retailers, and customers

- Within each company, the supply chain includes all functions involved in fulfilling a customer need/want
  - Product development, marketing, operations, distribution, finance, customer service and others

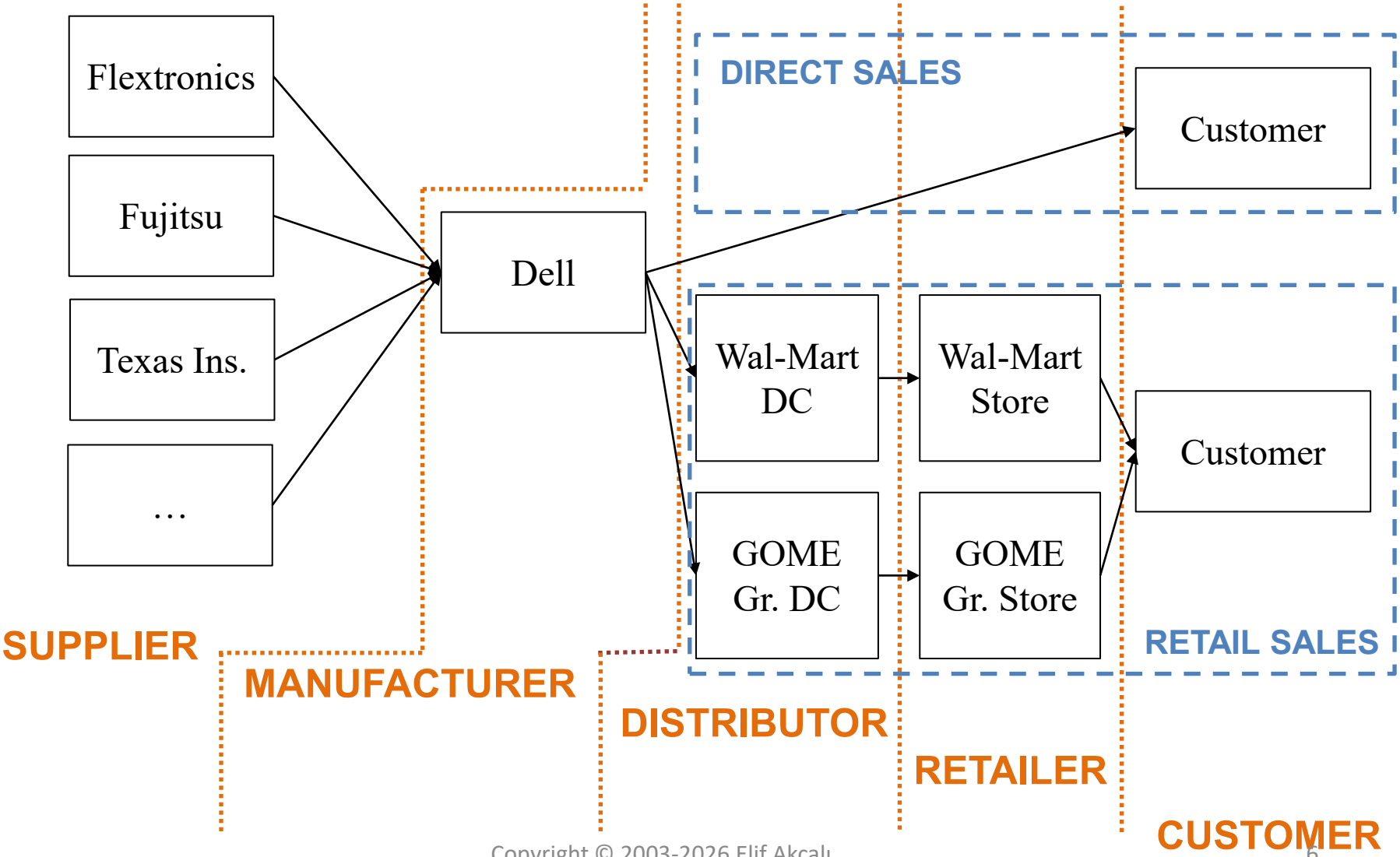
Copyright © 2003-2026 Elif Akçalı

What is a supply chain?

- Comprises of multiple types of flow
  - Information, product, and financial
- Typically includes multiple stages
  - Customers, retailers, distributors, manufacturers and (component or raw material) suppliers
- But…
  - Not always “chains” but “networks” or “webs”
  - Not all stages may not be present in all supply chains (e.g., Dell)
  - Not the same for different customers (e.g., Dell)

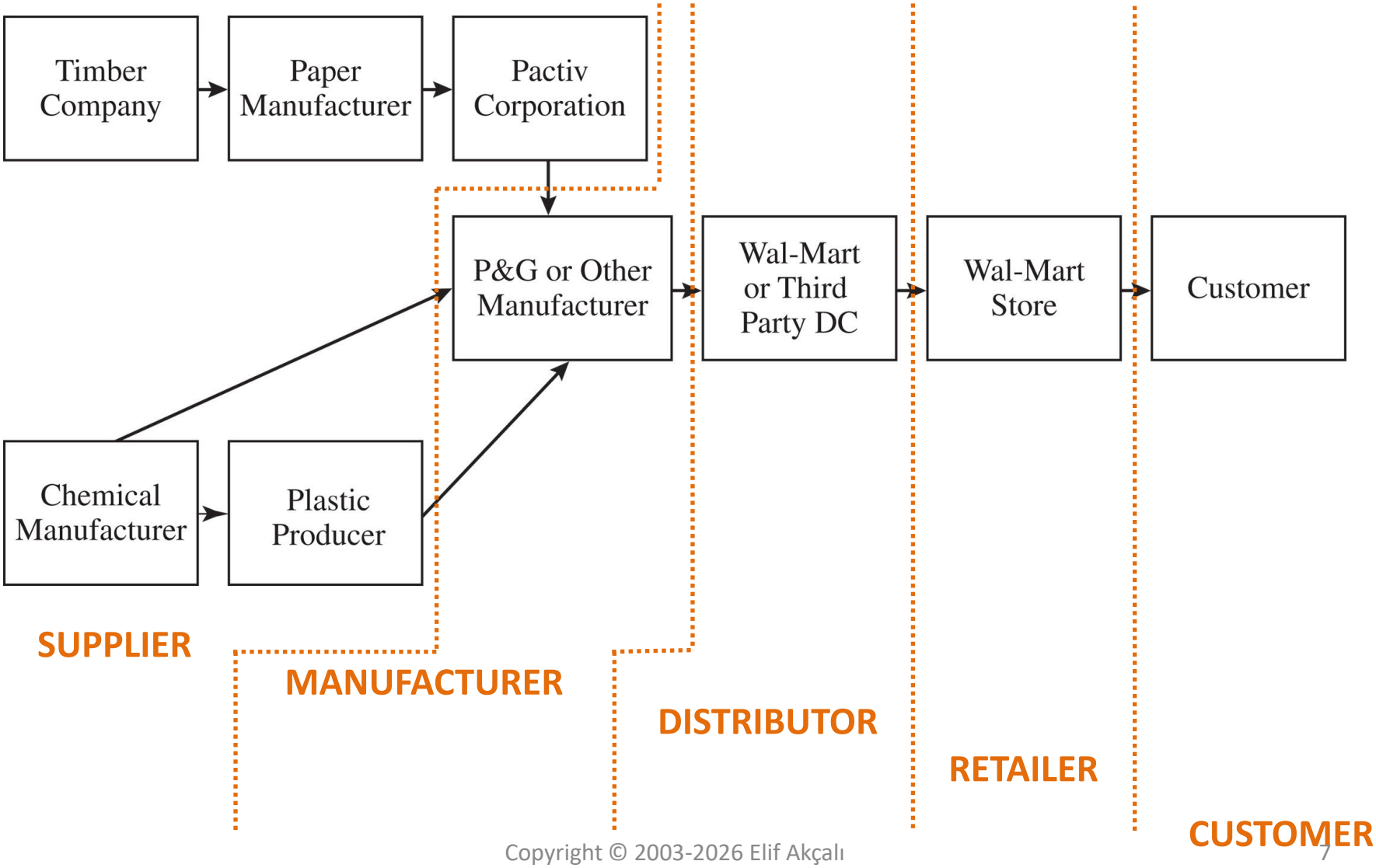
Copyright © 2003-2026 Elif Akçalı

Example: A computer supply chain



Copyright © 2003-2026 Elif Akçalı

Example: A detergent supply chain



Copyright © 2003-2026 Elif Akçalı

Are supply chain decisions important?

- Appropriateness of design, planning and operation of supply chain decisions dictate the success of a supply chain

- *The Good, the Bad and the Ugly*
  - Wal-Mart: success in information sharing and collaboration with suppliers
  - Borders: failure to adapt to changes in the environment
  - Webvan: inability to design effective supply chains

Competitiveness requires an ability to adapt to changing technology and customer needs and desires

Copyright © 2003-2026 Elif Akçalı

What are supply chain decisions?

Problem areas and typical questions that we try to answer for a supply chain include:

- **Purchasing:** what, how much, when, where
- **Production:** what, how much, when, where
- **Inventory:** what, how much, when, where
- **Workforce:** which skills, how many, when, where
- **Facilities:** what kind, what capacity, when, where
- **Distribution:** what, how much, how, when, where

Copyright © 2003-2026 Elif Akçalı

Objective of a supply chain

- Maximize overall value (or surplus) generated given by

CUSTOMER VALUE	-	SUPPLY CHAIN COST	=	SUPPLY CHAIN SURPLUS

- What the customer is willing to pay minus the total cost that all the agents in the entire supply chain incurs

Copyright © 2003-2026 Elif Akçalı

How do you evaluate the financial performance of supply chain operations?

Optimization of the system that creates and delivers a company's products can be source of competitive advantage and improve the performance of the company.

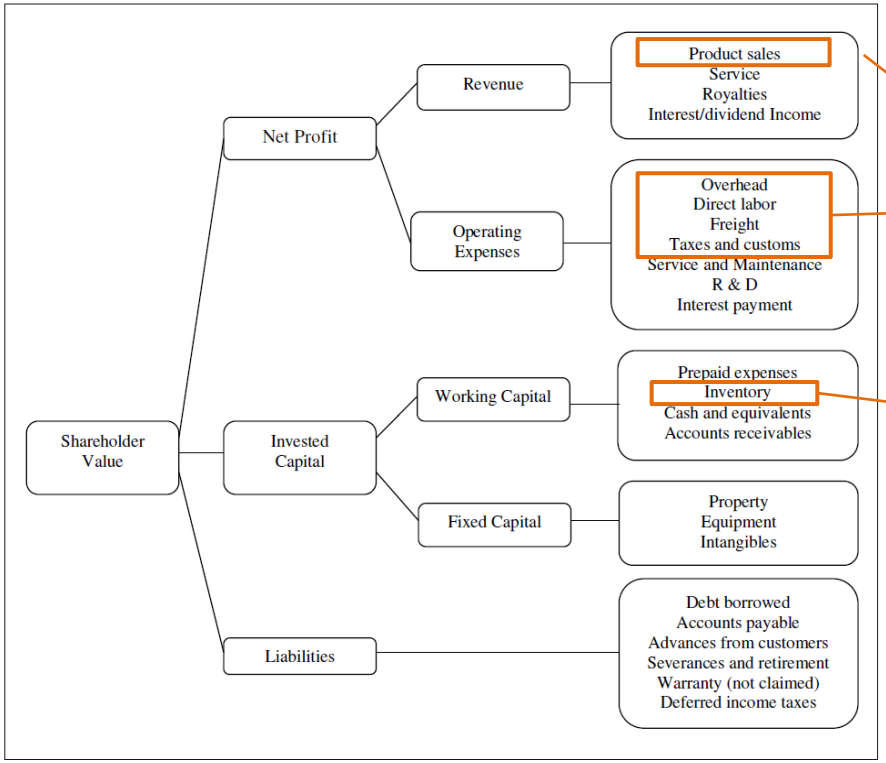
A performance measure: Return on investment (ROI), given by

$$ROI = \frac{\text{Net Profit}}{\text{Net Assets}},$$

provides an estimate of the financial benefit of the company on money that is spent.

Copyright © 2003-2026 Elif Akçalı

Components of shareholder value



Net profit
(numerator of ROI)

Net assets
(denominator of ROI)

1. **Supply chain operation decisions influence both the numerator and the denominator of ROI.**
2. **Improvement of supply chain operations can increase the value of the denominator and decrease the value of the denominator.**

Copyright © 2003-2026 Elif Akçalı